

Lecture 04B

Linear Maps

Range and Null Space

Let $S = \{(1,0,0), (0,2,3)\}$.

Which of the following vectors belong to $\text{span}(S)$?

Choose ALL correct answers

1. $(0,0,0)$ $\lambda = 0, \mu = 0$
2. $(1,2,3)$ $\lambda = 1, \mu = 1$
3. $(0,4,6)$ $\lambda = 0, \mu = 2$
4. $(0,1048576,0)$ No such λ, μ

$$\text{span}(S) = \left\{ \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} : \lambda, \mu \in \mathbb{R} \right\}$$

We set $(0, 1048576, 0)$

$$= \lambda (1, 0, 0)$$

$$+ \mu (0, 2, 3)$$

and we find no solⁿ

$\therefore$ does not belong to $\text{span}(S)$.

Lecture 04B

- Linear Map  *special functions*
 - Definition, Examples, Properties
- Null Space and Range
 - Special Subspaces Associated with a Linear Map

Linear Map AKA linear transformation

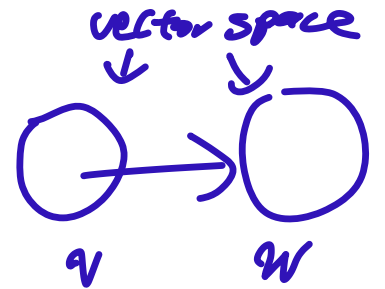
Let $\mathcal{V}$, $\mathcal{W}$ be vector spaces. The function $T : \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- **(Additivity).**

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in \mathcal{V}.$$

- **(Homogeneity).**

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$



- linear map is a special function

Example. Consider the linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} \right\}$$

$$T(x, y, z) = (2x - 3y + 3z, 7x + 5y - 6z)^T.$$

If $T(1, 0, 0) = (2, 7)^T$ and $T(0, 2, 3) = (3, -8)^T$, then what is $T(1, 2, 3)$? What is $T(0, 4, 6)$?

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T: \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- **(Additivity).**

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in \mathcal{V}.$$

- **(Homogeneity).**

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

(additive)

$$\frac{1}{1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \frac{1}{1} \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} \right)$$

$$= \frac{1}{1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{1} \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 7 \\ -1 \end{bmatrix} + \begin{bmatrix} 3 \\ -8 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$$

(Homogeneity)

$$\frac{1}{1} \begin{bmatrix} 0 \\ 4 \\ 6 \end{bmatrix} = \frac{1}{1} (2 \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}) = 2 \frac{1}{1} \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ -8 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -16 \\ -2 \end{bmatrix}$$

Example. Consider the function $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with

$$T(x, y, z) = (2x - 3y + 3z, 7x + 5y - 6z)^T.$$

Show that T is a linear map.

$$= \begin{bmatrix} 2 & -3 & 3 \\ 7 & 5 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T: \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- (Additivity).

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in \mathcal{V}.$$

- (Homogeneity).

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

(additive) For all $(x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3$

$$\frac{1}{T} [(x_1, y_1, z_1) + (x_2, y_2, z_2)] = \frac{1}{T} [(x_1 + x_2, y_1 + y_2, z_1 + z_2)]$$

$$= \begin{bmatrix} 2(x_1 + x_2) - 3(y_1 + y_2) + 3(z_1 + z_2) \\ 7(x_1 + x_2) + 5(y_1 + y_2) - 6(z_1 + z_2) \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1 - 3y_1 + 3z_1 \\ 7x_1 + 5y_1 - 6z_1 \end{bmatrix} + \begin{bmatrix} 2x_2 - 3y_2 + 3z_2 \\ 7x_2 + 5y_2 - 6z_2 \end{bmatrix}$$

$$= \frac{1}{T} (x_1, y_1, z_1) + \frac{1}{T} (x_2, y_2, z_2)$$

(Homogeneity) $\rightarrow$ Exercise

Proposition. (Matrix Multiplication is a Linear Map)

Let A be an $m \times n$ matrix. If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined by $T(v) = Av$, then T is a linear map.

$$\begin{matrix} m \\ \{ \overset{n}{A} \} \end{matrix} \cdot \begin{matrix} n \\ \{ x \} \end{matrix} = \begin{matrix} m \\ \{ \dots \} \end{matrix} \quad \begin{matrix} \mathbb{R}^n & \xrightarrow{T} & \mathbb{R}^m \end{matrix}$$

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T: \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- **(Additivity).**

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in \mathcal{V}.$$

- **(Homogeneity).**

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

(Additivity) For $u, v \in \mathbb{R}^n$, we have

$$T(u + v) = A(u + v) = Au + Av = T(u) + T(v)$$

(Homogeneity) For $u \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$, we have

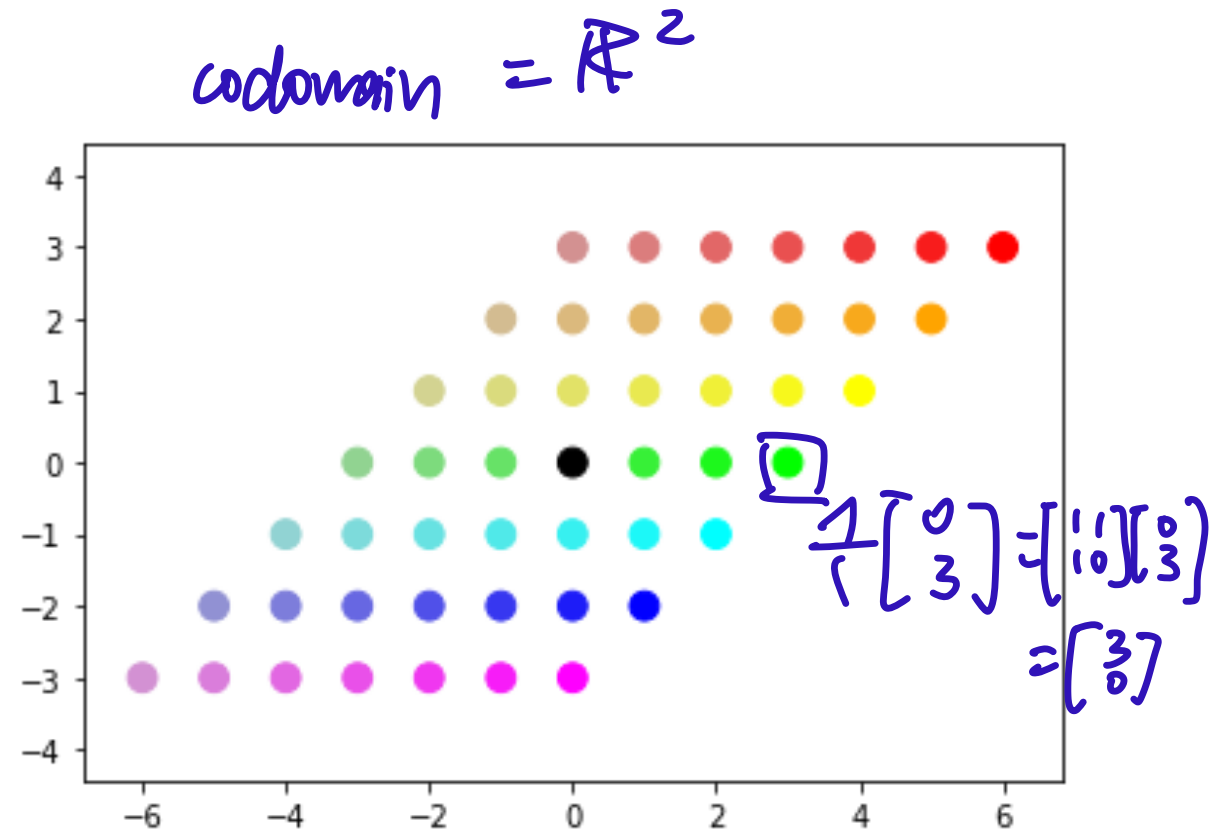
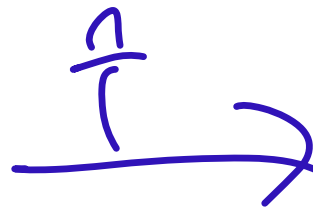
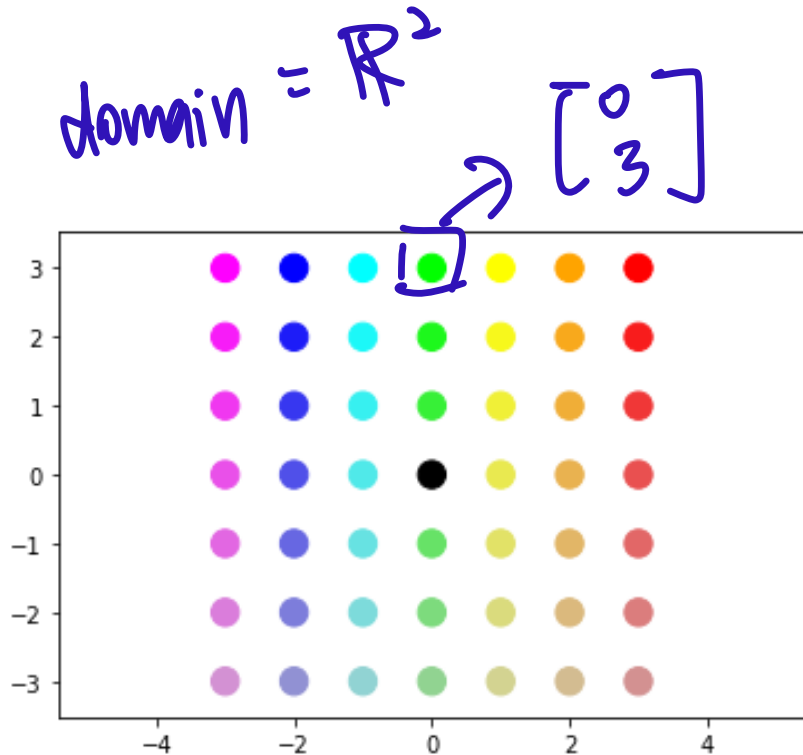
$$T(\lambda u) = A(\lambda u) = \lambda(Au) = \lambda T(u)$$

Proposition. (Matrix Multiplication is a Linear Map)

Let A be a 2×2 -matrix.

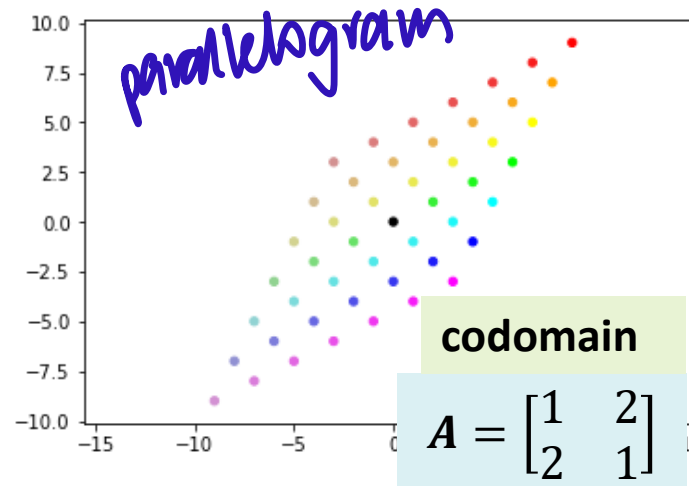
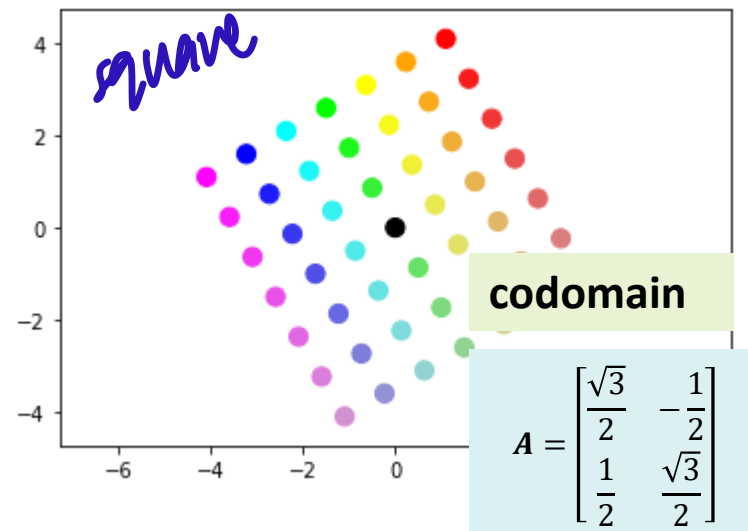
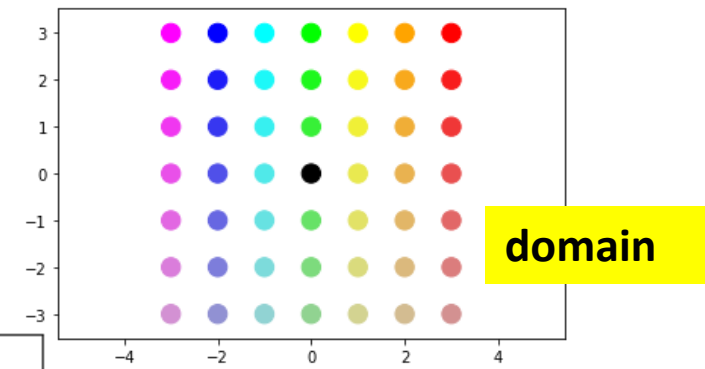
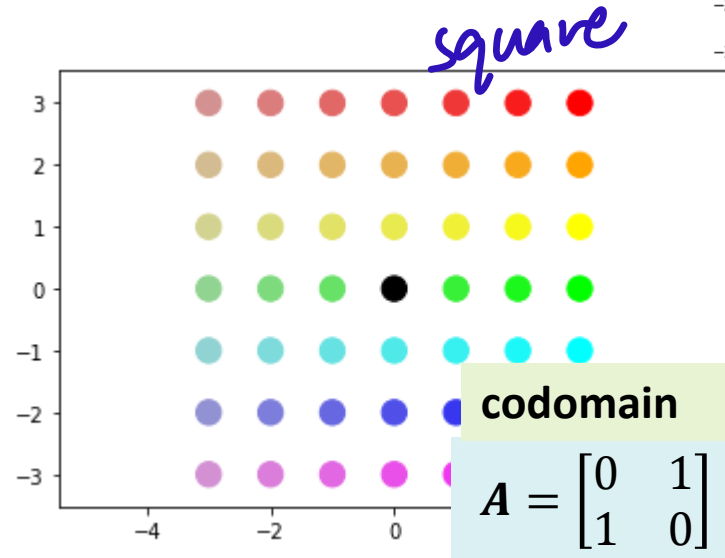
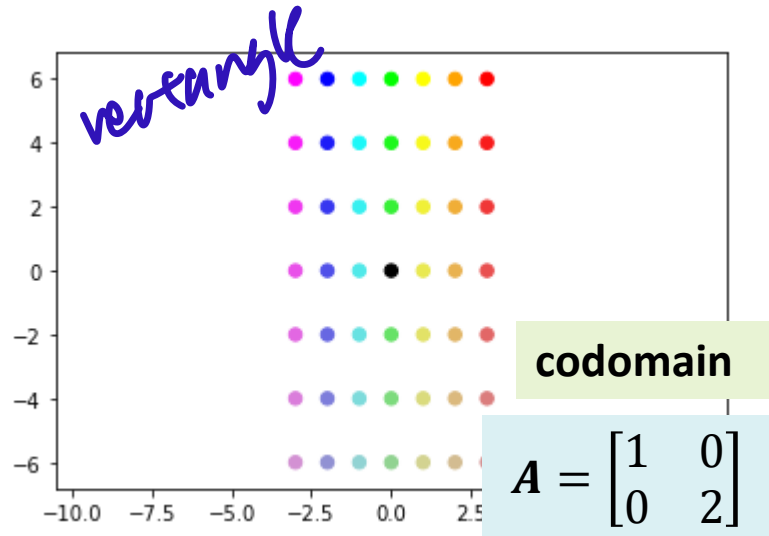
$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by $T(\mathbf{v}) = A\mathbf{v}$, then T is a linear map.



Sincere apologies to the colorblind.

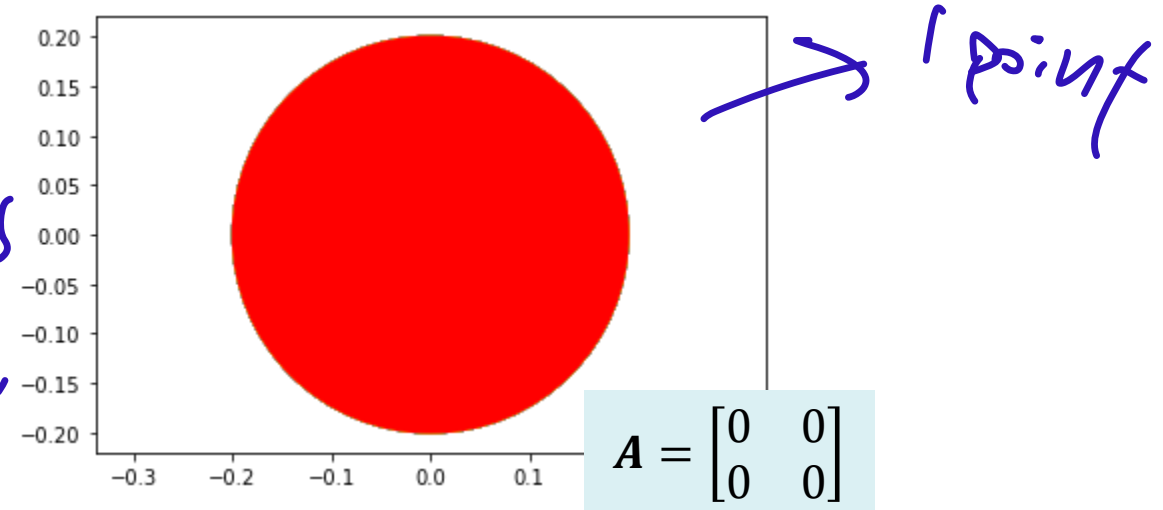
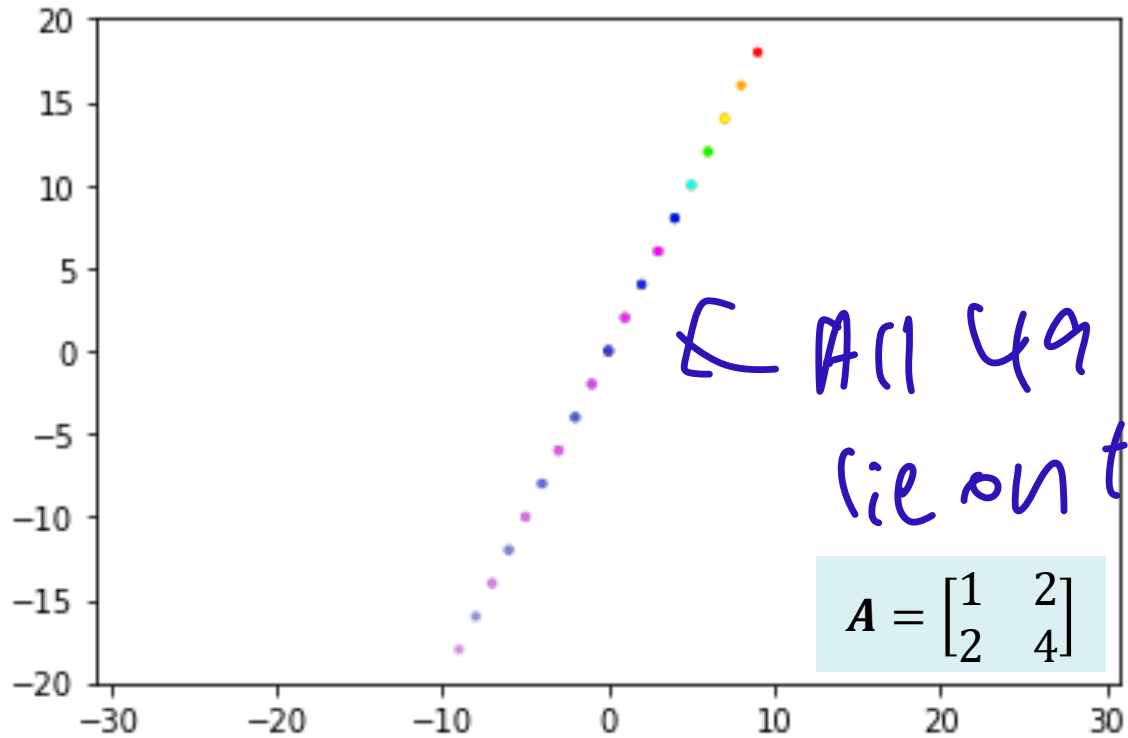
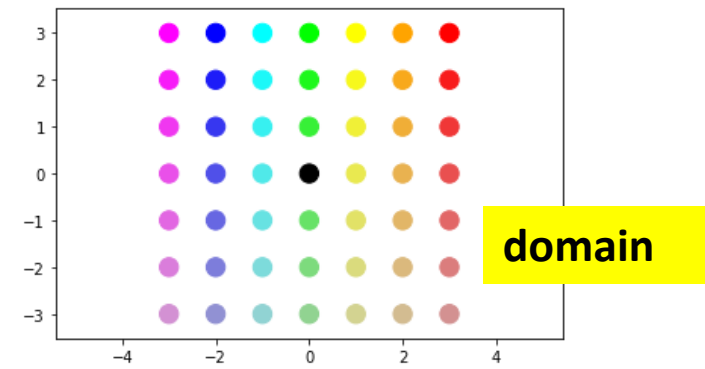
More Examples of A



Things we observe :

- Resulting shapes are "nice"
- SQUARE $\rightarrow$ parallelogram
- When will the shape be a square/rectangle?

More Examples of **A**



Proposition. (Differentiation of Polynomials)

If $D: \mathcal{P}_d(\mathbb{R}) \rightarrow \mathcal{P}_d(\mathbb{R})$ is defined by $D(f) = f'$, then D is a linear map.

→ differentiation

$$\text{ie } f = a_0 + a_1x + \dots + a_dx^d$$

$$\text{then } D(f) = a_1 + 2a_2x + \dots + da_dx^{d-1}$$

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T: \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- (Additivity).

$$T(u + v) = T(u) + T(v) \text{ for all } u, v \in \mathcal{V}.$$

- (Homogeneity).

$$T(\lambda u) = \lambda T(u) \text{ for all } u \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

(Additive) For $f = a_0 + a_1x + \dots + a_dx^d$

$$g = b_0 + b_1x + \dots + b_dx^d$$

$$D(f+g) = D(f+g) = D((a_0+b_0) + \dots + (a_d+b_d)x^d)$$

$$a_1 + 2a_2x + \dots + da_dx^{d-1}$$

$$+ b_1 + 2b_2x + \dots + db_dx^{d-1}$$

$$= D(f) + D(g)$$

$\therefore$ Differentiation is additive.

Homogeneity $\Rightarrow$ DIY

Proposition. (Differentiation is a Linear Map)

Let $\mathcal{D}$ be the set of all differentiable functions from $\mathbb{R}$ to $\mathbb{R}$.

If $D: \mathcal{D} \rightarrow \mathcal{F}(\mathbb{R}, \mathbb{R})$ is defined by $D(f) = f'$, then D is a linear map.

DIFFERENTIATION RULES

General Formulas

Assume u and v are differentiable functions of x .

Constant: $\frac{d}{dx}(c) = 0$

Sum: $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$

Difference: $\frac{d}{dx}(u - v) = \frac{du}{dx} - \frac{dv}{dx}$

Constant Multiple: $\frac{d}{dx}(cu) = c \frac{du}{dx}$

Product: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Quotient: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Power: $\frac{d}{dx}x^n = nx^{n-1}$

Chain Rule: $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$

Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \quad \frac{d}{dx}(\cot x) = -\csc^2 x$$

Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

Hyperbolic Functions

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\frac{d}{dx}(\coth x) = -\operatorname{csch}^2 x$$

Inverse Hyperbolic Functions

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

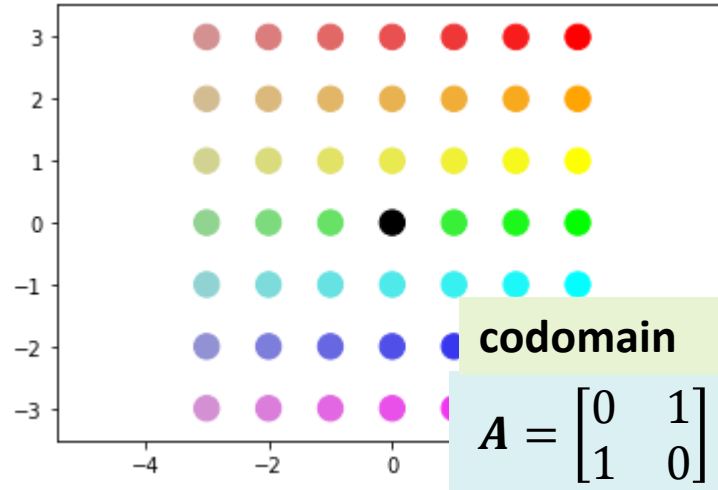
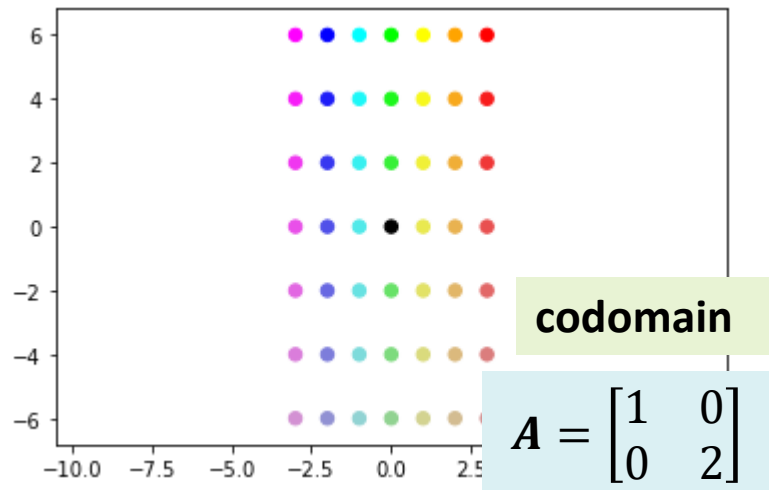
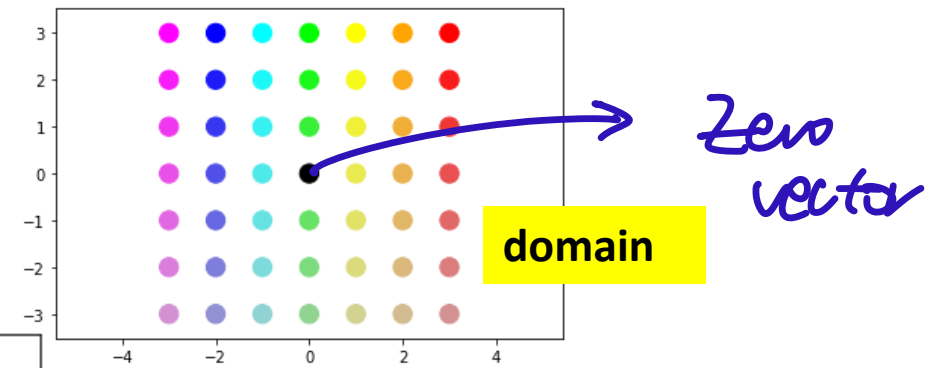
* Proof is omitted

Find $\frac{d}{dx} e^{3x} (\sin x + 2 \cos x) \dots = \lambda$

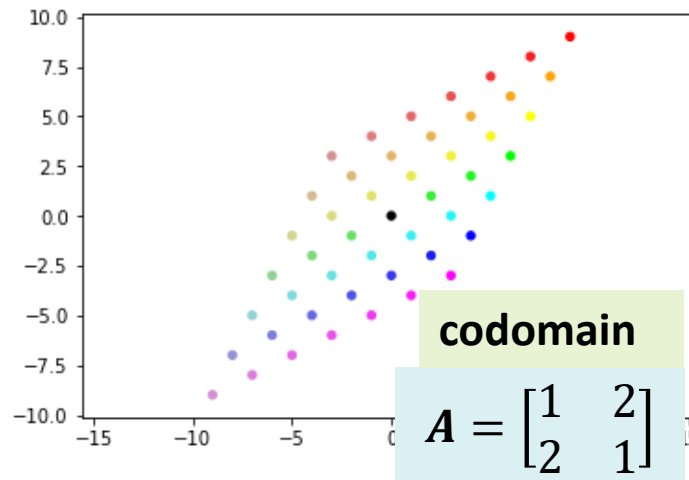
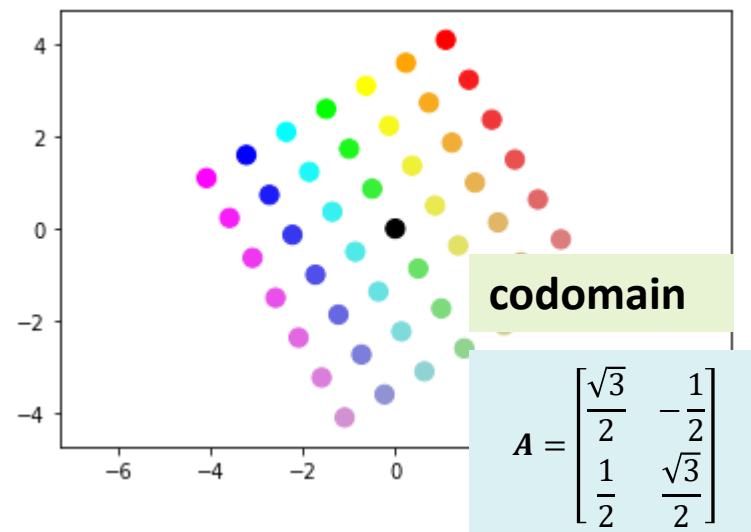
What we will learn in next three weeks

- Examples of Linear Maps
 - Matrix Multiplication Map
 - Zero Map
 - Identity Map
 - Differentiation Map
- Properties of Linear Map
 - $T(\mathbf{0})$ is always...
 - $S + T$ is also linear
 - μT is also linear
- Special Subspaces of T
 - Nullspace
 - Range
 - Relation to injectivity, surjectivity and isomorphism
 - Rank-Nullity Theorem
- Matrix Representation of T

$T(\mathbf{0})$ is always...



black dot always @ same position



Proposition. (Zero maps to zero)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be any linear map.

Then $T(\mathbf{0}) =$

HIDDEN

0 $\rightarrow$ zero vector

Many ways to prove this!

Recall: $\underline{0} = \underline{0} + \underline{0}$

$$\underline{0} = 0 \cdot \underline{u}$$

$$\underline{0} = \underline{u} - \underline{u}$$

any vector

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T: \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- (Additivity).

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \text{ for all } \mathbf{u}, \mathbf{v} \in \mathcal{V}.$$

- (Homogeneity).

$$T(\lambda \mathbf{u}) = \lambda T(\mathbf{u}) \text{ for all } \mathbf{u} \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

Pf

$$\text{So, } T(\underline{0}) = T(\underline{0} + \underline{0}) = T(\underline{0}) + T(\underline{0})$$

We can subtract $T(\underline{0})$ from both sides.

$$\text{LHS} = T(\underline{0}) - T(\underline{0}) = \underline{0}$$

Since LHS=RHS
we have

$$\text{RHS} = T(\underline{0}) + (T(\underline{0}) - T(\underline{0})) \therefore T(\underline{0}) = \underline{0}$$

$$= T(\underline{0}) + \underline{0}$$

$$= T(\underline{0})$$

Linear Maps are Special Functions

Recall: When we learnt about functions, we did the following:

- Define function f .
- Is f surjective?
- Is f injective?
- Is f bijective?

} when f is linear, performing these tasks will be easier.

Let $\mathcal{V}, \mathcal{W}$ be vector spaces.

The function $T : \mathcal{V} \rightarrow \mathcal{W}$ is a linear map if the following two conditions hold:

- **(Additivity).**

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \text{ for all } \mathbf{u}, \mathbf{v} \in \mathcal{V}.$$

- **(Homogeneity).**

$$T(\lambda \mathbf{u}) = \lambda T(\mathbf{u}) \text{ for all } \mathbf{u} \in \mathcal{V} \text{ and } \lambda \in \mathbb{F}.$$

Linear Maps

- Defining a Linear Map
- Range and Surjectivity
- Null space and Injectivity

Examples of Linear Maps

	domain	codomain	Linear Map	Nullspace	Range
zero map	$\mathcal{V}$	$\mathcal{W}$	$\mathbf{Z}(v) = \mathbf{0}$	$\mathcal{W}$	$\mathcal{W}$
identity map	$\mathcal{V}$	$\mathcal{V}$	$\mathbf{I}(v) = v$	$\mathcal{W}$	$\mathcal{W}$
matrix multiplication	$\mathbb{R}^3$	$\mathbb{R}$	$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $x + 2y + 3z$	today	today
differentiation map	$\mathcal{P}_3(\mathbb{R})$	$\mathcal{P}_3(\mathbb{R})$	$\mathbf{D}(f) = f'$	today	today

Describing a set (in general)

- List everything

Describing a vector space

- Use a basis!

Defining a function (in general)

→ Define $f(x)$ for all x in X

→ domain

Defining a linear map

→ use a basis

→ Define $f(x)$ for x in a basis

Theorem. (Linear Map is Uniquely Defined by Basis of Domain)

Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$. Let $w_1, \dots, w_n$ vectors in $\mathcal{W}$.
→ domain

Then there exists a *unique linear map* $T: \mathcal{V} \rightarrow \mathcal{W}$ such that

$$T(v_i) = w_i \text{ for all } i = 1, \dots, n.$$

"Two parts" to this theo rem

Part 1 : Linear map is unique *→ use this part to describe range*

Part 2: Linear map exists *→ colab discuss what this means*

Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 1 rephrased)
Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$.
If we know $T(v_i)$ for all $i = 1, \dots, n$, we can determine $T(v)$ for all $v \in \mathcal{V}$.

Example. Consider $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ a basis for $\mathbb{R}^2$. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear map such that $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix}$ and $T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 7 \\ 8 \\ -3 \end{bmatrix}$.
Then what is $T\left(\begin{bmatrix} 3 \\ 14 \end{bmatrix}\right)$?

Write $\begin{bmatrix} 3 \\ 14 \end{bmatrix}$ as a lin. combination of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
Solve $\begin{bmatrix} 3 \\ 14 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 $\Rightarrow \lambda_1 = -4, \lambda_2 = 7$

$T\left(\begin{bmatrix} 3 \\ 14 \end{bmatrix}\right) = T\left(-4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 7 \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$
 $= -4 T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + 7 T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right)$ - works for any $T\left[\begin{bmatrix} x \\ y \end{bmatrix}\right]$
 $= -4 \begin{bmatrix} -4 \\ 0 \\ 1 \end{bmatrix} + 7 \begin{bmatrix} 7 \\ 8 \\ -3 \end{bmatrix} = \begin{bmatrix} 28 \\ 56 \\ -25 \end{bmatrix}$ - Question: why does this work?
see Part 1 for the proof $\downarrow$

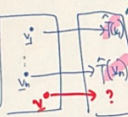
Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 1 rephrased)

Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$.

If we know $T(v_i)$ for all $i = 1, \dots, n$, we can determine $T(v)$ for all $v \in \mathcal{V}$.

$$\left. \begin{array}{l} v_1 \rightarrow T(v_1) \\ \vdots \\ v_n \rightarrow T(v_n) \end{array} \right\} \text{I will be able to} \\ \text{determine } \hat{T}(v) \text{ for all } v.$$

Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 1 rephrased)
Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$.
If we know $T(v_i)$ for all $i = 1, \dots, n$, we can determine $T(v)$ for all $v \in \mathcal{V}$.


PF Let $v \in \mathcal{V}$. Since $[v_1, \dots, v_n]$ is a basis,
by Unique Representation Theorem,
we can write $v = \lambda_1 v_1 + \dots + \lambda_n v_n$.
$$\hat{T}(v) = \hat{T}(\lambda_1 v_1 + \dots + \lambda_n v_n) = \hat{T}(\lambda_1 v_1) + \dots + \hat{T}(\lambda_n v_n)$$

$$= \lambda_1 \hat{T}(v_1) + \dots + \lambda_n \hat{T}(v_n)$$

So, we can determine $\hat{T}(v)$

Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 2 rephrased)
Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$. Let $w_1, \dots, w_n$ vectors in $\mathcal{W}$.
Then there exists a linear map $T: \mathcal{V} \rightarrow \mathcal{W}$ such that
 $T(v_i) = w_i$ for all $i = 1, \dots, n$.

① Define $\hat{T}$
② Show that $\hat{T}$ is linear.
③ Show that $\hat{T}(v_i) = w_i$ for all i .
Same as part 1.
Let $v \in \mathcal{V}$. Since $[v_1, \dots, v_n]$ is a basis for $\mathcal{V}$, we can write
 $v = \lambda_1 v_1 + \dots + \lambda_n v_n$.
Then $\hat{T}(v) = \lambda_1 w_1 + \dots + \lambda_n w_n$

Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 2 rephrased)
Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$. Let $w_1, \dots, w_n$ vectors in $\mathcal{W}$.
Then there exists a linear map $T: \mathcal{V} \rightarrow \mathcal{W}$ such that
 $T(v_i) = w_i$ for all $i = 1, \dots, n$.

② Show that $\hat{T}$ is linear.
i.e. $\hat{T}$ is additive
 $\hat{T}$ is homogeneous
For $u, v \in \mathcal{V}$,
By Unique Rep, $u = \lambda_1 v_1 + \dots + \lambda_n v_n$ and $v = \mu_1 v_1 + \dots + \mu_n v_n$
$$\hat{T}(u+v) = \hat{T}(\lambda_1 v_1 + \dots + \lambda_n v_n + \mu_1 v_1 + \dots + \mu_n v_n)$$

$$= \hat{T}((\lambda_1 + \mu_1) v_1 + \dots + (\lambda_n + \mu_n) v_n)$$

$$= (\lambda_1 + \mu_1) w_1 + \dots + (\lambda_n + \mu_n) w_n$$

Check that
$$\hat{T}(u) + \hat{T}(v) = \lambda_1 w_1 + \dots + \lambda_n w_n + \mu_1 w_1 + \dots + \mu_n w_n$$

$$= (\lambda_1 + \mu_1) w_1 + \dots + (\lambda_n + \mu_n) w_n$$

$$= \hat{T}(u+v)$$

Theorem. (Linear Map is Uniquely Defined by Basis of Domain – Part 2 rephrased)
Let $[v_1, \dots, v_n]$ be a basis for $\mathcal{V}$. Let $w_1, \dots, w_n$ vectors in $\mathcal{W}$.
Then there exists a linear map $T: \mathcal{V} \rightarrow \mathcal{W}$ such that
 $T(v_i) = w_i$ for all $i = 1, \dots, n$.

③ $\hat{T}(v_i) = w_i$ for all i .
Write v_i as a linear combination of v_j 's.
$$v_i = 0 \cdot v_1 + \dots + 1 \cdot v_i + \dots + 0 \cdot v_n$$

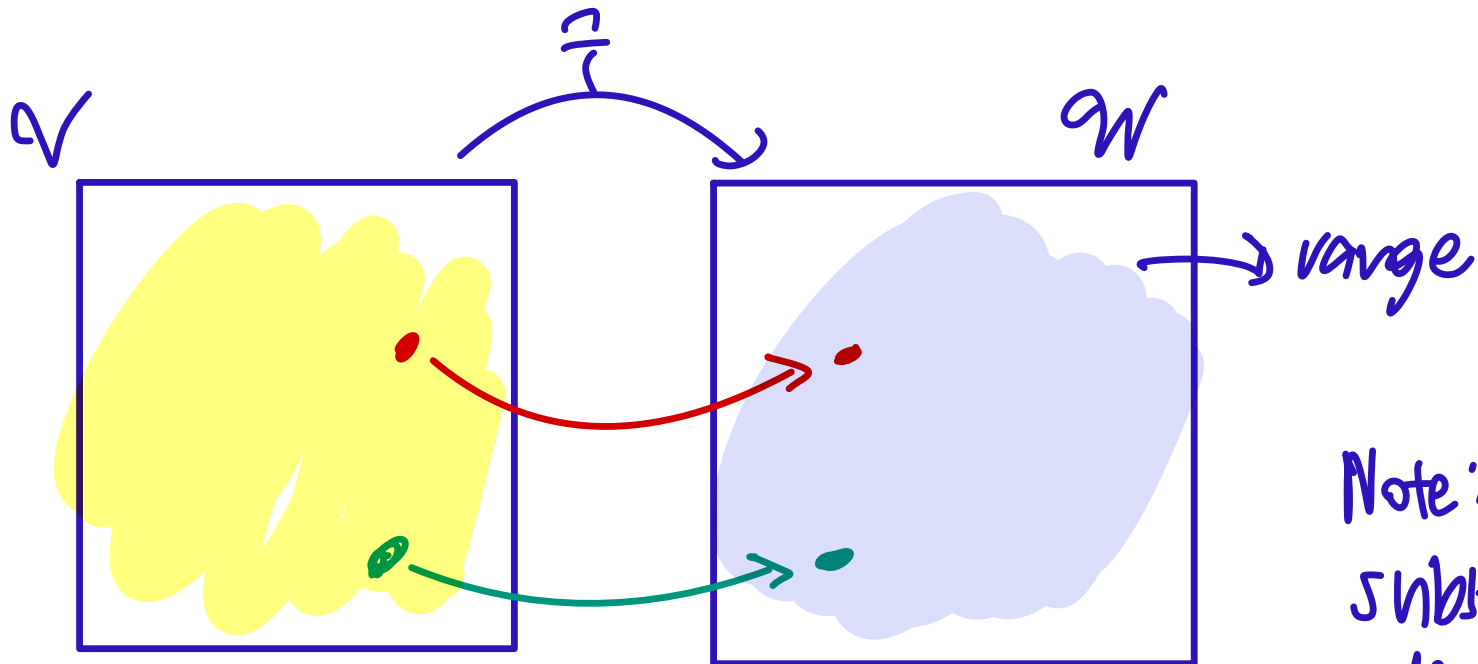
By Unique Rep. Theorem, since v_j 's form a basis,
this is the only linear combination.
$$\therefore \hat{T}(v_i) = 0 w_1 + \dots + 1 w_i + \dots + 0 w_n = w_i. \quad \square$$

Range

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

The range of T , denoted by $\text{range}(T)$, is the subset of $\mathcal{W}$ consisting of vectors that are mapped by T from vectors in $\mathcal{V}$.

$$\text{range}(T) = \underline{\{T(v) : v \in \mathcal{V}\}}.$$



Note: $\text{range}(T)$ is a subset of the codomain

Proposition. (Range is a vector space)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

Then range of T is a subspace of $\mathcal{W}$.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map

$$\text{range}(T) = \{T(v) : v \in \mathcal{V}\}.$$

• 0 belongs to $\text{range}(T)$ because $T(0) = 0$.

(closed under $\oplus$) For $w_1, w_2 \in \text{range}(T)$

i.e. there exist x_1, x_2 s.t. $T(x_1) = w_1$ & $T(x_2) = w_2$

$$\text{Now, } T(x_1 + x_2) = T(x_1) + T(x_2) = w_1 + w_2$$

$$\Rightarrow w_1 + w_2 \in \text{range}(T)$$

• (closed under $\otimes$) For $w \in \text{range}(T)$, $\lambda \in \mathbb{F}$,

i.e. there exists $x \in \mathcal{V}$ s.t. $T(x) = w$.

$$T(\lambda x) = \lambda T(x) = \lambda w$$

$$\Rightarrow \lambda w \in \text{range}(T)$$

T is linear/
homogeneous

T is linear/additive

How to determine the range?

Theorem. (Linear Map is Uniquely Defined by Basis of Domain)

Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for $\mathcal{V}$. Let $\mathbf{w}_1, \dots, \mathbf{w}_n$ vectors in $\mathcal{W}$.

Then there exists a *unique linear map* $T: \mathcal{V} \rightarrow \mathcal{W}$ such that

$$T(\mathbf{v}_i) = \mathbf{w}_i \text{ for all } i = 1, \dots, n.$$

Proposition. (Range is Uniquely Defined by Basis of Domain)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map and let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for $\mathcal{V}$.

Then

$$\text{range}(T) = \text{span}(\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_n)\}).$$

Example. Consider the linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ with

$$T(x, y, z) = x + 2y + 3z = [1 \quad 2 \quad 3] \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Determine the range of T .

↓
matrix

↓
3-tuple

domain
 $\mathbb{R}^3$ has standard basis
 $= \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map

$$\text{range}(T) = \{T(v) : v \in \mathcal{V}\}.$$

$$\begin{aligned} \text{range}\left(\frac{1}{1}\right) &= \text{span} \left\{ \frac{1}{1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \frac{1}{1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \\ &= \text{span} \{1, 2, 3\} = \mathbb{R} \end{aligned}$$

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

If $\{v_1, \dots, v_n\}$ is a basis for $\mathcal{V}$, then

$$\text{range}(T) = \text{span}(T(v_1), T(v_2), \dots, T(v_n)).$$

Example. Consider the linear map $D: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ with

$$D(f) = f'.$$

Determine the range of D .

Domain $\mathcal{P}_3(\mathbb{R})$ has
standard basis
 $[1, x, x^2, x^3]$

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map

$$\text{range}(T) = \{T(v) : v \in \mathcal{V}\}.$$

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

If $\{v_1, \dots, v_n\}$ is a basis for $\mathcal{V}$, then

$$\text{range}(T) = \text{span}(T(v_1), T(v_2), \dots, T(v_n)).$$

$$\begin{aligned} \text{range}(\hat{D}) &= \text{span}(\hat{D}(1), \hat{D}(x), \hat{D}(x^2), \hat{D}(x^3)) \\ &= \text{span}(\cancel{0}, 1, 2x, 3x^2) \\ &= \text{span}(1, x, x^2) = \mathcal{P}_2(\mathbb{R}) \end{aligned}$$

Ⓢ Range may not be the
codomain.

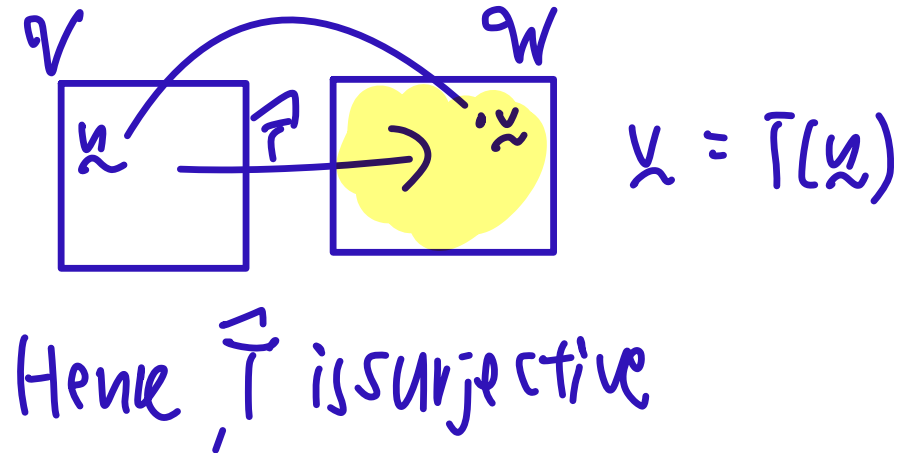
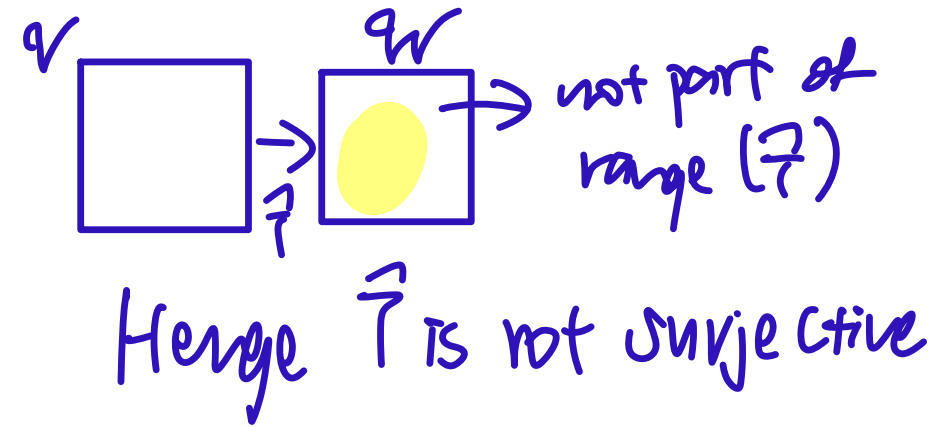
$$\neq \mathcal{P}_3(\mathbb{R})$$

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.
 T is surjective if $\text{range}(T) = \mathcal{W}$.

Proposition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

If $\{v_1, \dots, v_n\}$ is a basis for $\mathcal{V}$, then

$$\text{range}(T) = \text{span}(T(v_1), T(v_2), \dots, T(v_n)).$$



To check that $\hat{T}$ is surjective
 — we compute $\text{range}(\hat{T})$ (which is more mechanical now)

Examples of Linear Maps

domain	codomain	Linear Map	Range	Surjective?
$\mathbb{R}^3$	$\mathbb{R}$	$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$\mathbb{R}$	BLANK1 <i>yes</i>
$\mathcal{P}_3(\mathbb{R})$	$\mathcal{P}_3(\mathbb{R})$	$D(f) = f'$	$\mathcal{P}_2(\mathbb{R})$	BLANK2 <i>No</i>

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.
 T is surjective if $\text{range}(T) = \mathcal{W}$.

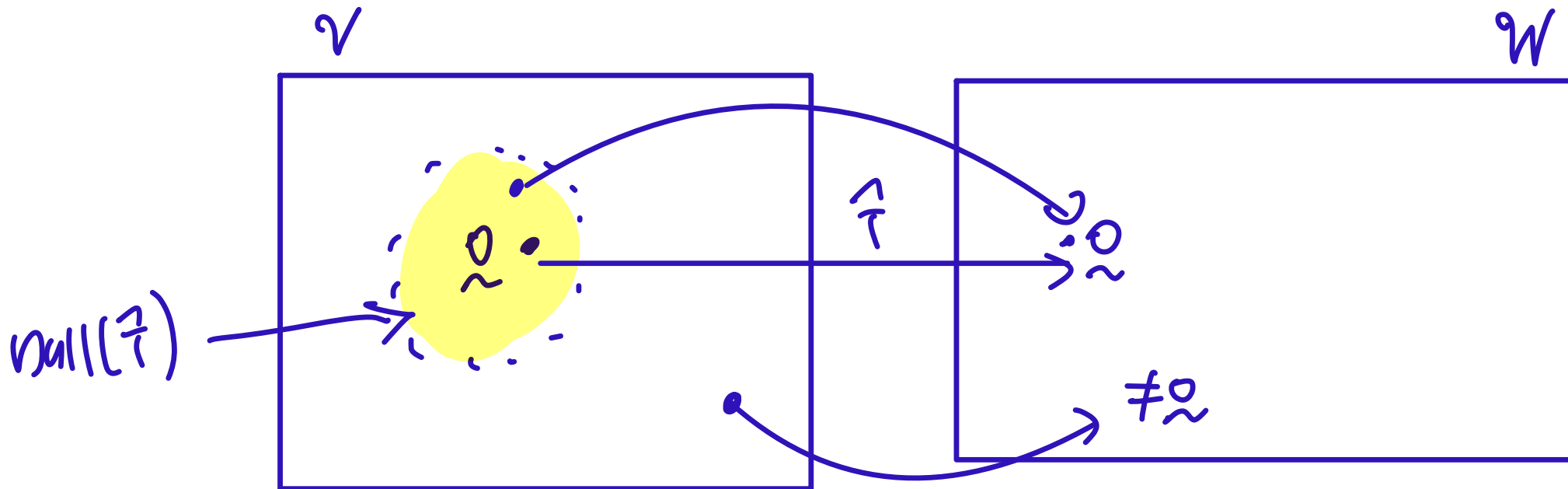
Is $\text{range}(T) = \mathcal{W}$?

Null Space

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map

The null space of $T: \mathcal{V} \rightarrow \mathcal{W}$, denoted by $\text{null}(T)$, is the subset of $\mathcal{V}$ consisting of vectors that T maps to $\mathbf{0}$:

$$\text{null}(T) = \{v \in \mathcal{V} : T(v) = \mathbf{0}\}.$$



Example. Consider the linear map $T: \mathbb{R}^3 \rightarrow \mathbb{R}$ with

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x + 2y + 3z.$$

Determine the null space of T .

When is

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = 0$$

$$\parallel \\ x + 2y + 3z$$

The null space of $T: \mathcal{V} \rightarrow \mathcal{W}$,

denoted by $\text{null}(T)$, is the

subset of $\mathcal{V}$ consisting of vectors

that T maps to $\mathbf{0}$:

$$\text{null}(T) = \{v \in \mathcal{V} : T(v) = \mathbf{0}\}.$$

$$\text{Solve : } x + 2y + 3z = 0$$

$$\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x = -2s - 3t, y = s, z = t, s, t \in \mathbb{R} \right\}$$

$$= \text{span} \left(\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right) \parallel \text{null}(T)$$

Example. Consider the linear map $D: \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ with

$$D(f) = f'.$$

Determine the null space of D .

Solve $D(f) = f' = 0$

zero
polynomial

Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$

$$f'(x) : a_1 + 2a_2x + 3a_3x^2 = 0$$

$$\Rightarrow a_1 = 0, 2a_2 = 0, 3a_3 = 0$$

$$\Rightarrow a_1 = a_2 = a_3 = 0 \quad f(x) = a_0$$

$$\therefore \text{null}(D) = \mathcal{P}_0(\mathbb{R}) = \{\text{all constant functions}\}$$

The null space of $T: \mathcal{V} \rightarrow \mathcal{W}$, denoted by $\text{null}(T)$, is the subset of $\mathcal{V}$ consisting of vectors that T maps to $\mathbf{0}$:

$$\text{null}(T) = \{v \in \mathcal{V} : T(v) = \mathbf{0}\}.$$

Proposition. (Null space is a vector space)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

Then the null space of T is a subspace of $\mathcal{V}$.

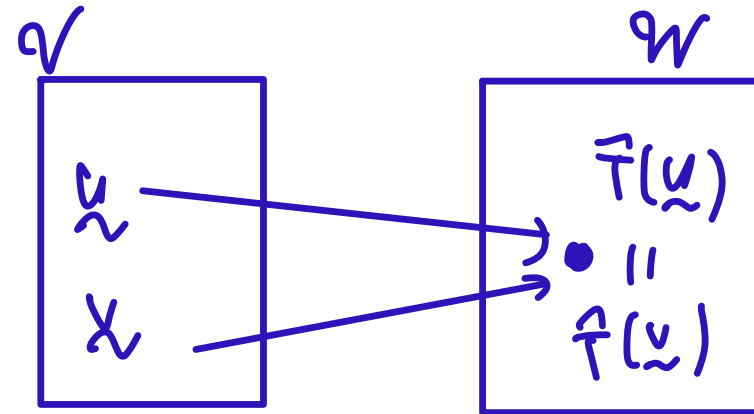
The null space of $T: \mathcal{V} \rightarrow \mathcal{W}$, denoted by $\text{null}(T)$, is the subset of $\mathcal{V}$ consisting of vectors that T maps to $\mathbf{0}$:
$$\text{null}(T) = \{v \in \mathcal{V} : T(v) = \mathbf{0}\}.$$

- $\mathbf{0}$ belongs to $\text{null}(T)$ because $T(\mathbf{0}) = \mathbf{0}$
- (closed under $\oplus$) For $u, v \in \text{null}(T)$
 $\Rightarrow T(u) = \mathbf{0}$ & $T(v) = \mathbf{0}$
 $\Rightarrow T(u+v) = T(u) + T(v) = \mathbf{0} + \mathbf{0} = \mathbf{0}$
 $\quad \quad \quad \downarrow T \text{ is linear}$
 $\Rightarrow u+v \in \text{null}(T)$
- (closed under $\otimes$) For $x \in \text{null}(T)$, $\lambda \in \mathbb{F}$,
 $\Rightarrow T(x) = \mathbf{0}$
 $\Rightarrow T(\lambda x) = \lambda T(x) = \lambda \mathbf{0} = \mathbf{0}$
 $\Rightarrow \lambda x \in \text{null}(T)$

Definition. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective if

$T(u) = T(v)$ implies that $u = v$.

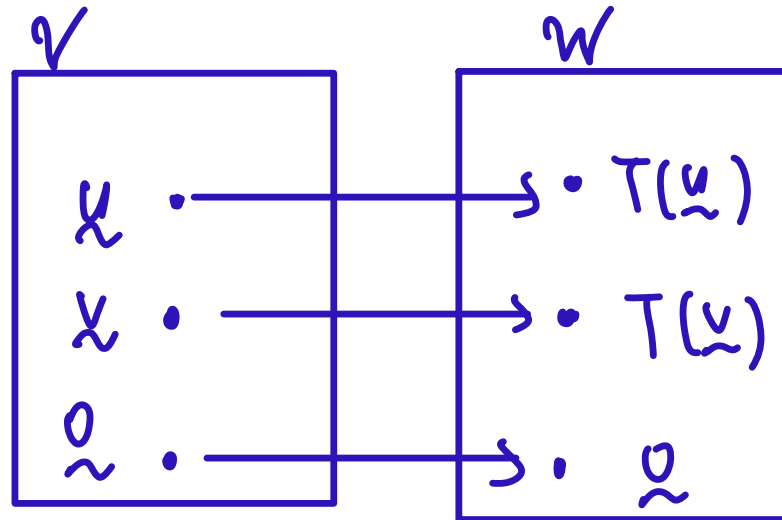


T is not injective
i.e. $\exists u \neq v$
s.t. $T(u) = T(v)$

Proposition. (Conditions for Injectivity)

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective if and only if $\text{null}(T) = \{0\}$.



T is injective
 $\text{null}(T) = \{0\}$

But Prop says that:

Just check 0 has exactly one preimage

Proof on colab

Proposition. (Conditions for Injectivity) Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective if and only if $\text{null}(T) = \{\mathbf{0}\}$.

T is injective if

$T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.

The null space of $T: \mathcal{V} \rightarrow \mathcal{W}$, denoted by $\text{null}(T)$, is the subset of $\mathcal{V}$ consisting of vectors that T maps to $\mathbf{0}$:

$$\text{null}(T) = \{\mathbf{v} \in \mathcal{V} : T(\mathbf{v}) = \mathbf{0}\}.$$

Proposition. (Conditions for Injectivity) Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective if and only if $\text{null}(T) = \{\mathbf{0}\}$.

T is injective if

$T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.

$\text{null}(T) = \{\mathbf{v} \in \mathcal{V} : T(\mathbf{v}) = \mathbf{0}\}$.

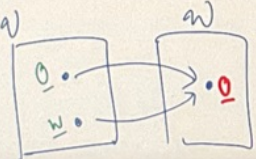
$(\Rightarrow)$ ~~If $\hat{T}$ is injective, then $\text{null}(\hat{T}) = \{\mathbf{0}\}$ (contrapositive)~~

WTS: if $\text{null}(\hat{T}) \neq \{\mathbf{0}\}$, then $\hat{T}$ is not injective.

WTS: if $\text{null}(\hat{T}) \neq \{\mathbf{0}\}$, then there exists $\mathbf{u} \neq \mathbf{v}$ s.t. $\hat{T}(\mathbf{u}) = \hat{T}(\mathbf{v})$

$\neg \text{null}(\hat{T}) \neq \{\mathbf{0}\} \Rightarrow$ there exists $\mathbf{w} \in \mathcal{V}$ s.t. $\mathbf{w} \neq \mathbf{0}$ and $\hat{T}(\mathbf{w}) = \mathbf{0}$.

- we can choose $\mathbf{u} = \mathbf{0}$, $\mathbf{v} = \mathbf{w}$.



Proposition. (Conditions for Injectivity) Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map.

T is injective if and only if $\text{null}(T) = \{\mathbf{0}\}$.

T is injective if

$T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.

$\text{null}(T) = \{\mathbf{v} \in \mathcal{V} : T(\mathbf{v}) = \mathbf{0}\}$.

$(\Rightarrow)$ WTS: if $\text{null}(\hat{T}) \neq \{\mathbf{0}\}$, then $\hat{T}$ is not injective.

if $\text{null}(\hat{T}) \neq \{\mathbf{0}\}$,

there exists $\mathbf{w} \neq \mathbf{0}$ s.t. $\hat{T}(\mathbf{w}) = \mathbf{0} = \hat{T}(\mathbf{0})$.

In other words, we have distinct vectors $\mathbf{w}, \mathbf{0} \in \mathcal{V}$ s.t. their images are both zero.

$\therefore \hat{T}$ is not injective.

Examples of Linear Maps

domain	codomain	Linear Map	Null space	Injective?
$\mathbb{R}^3$	$\mathbb{R}$	$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$	$\text{span}\left(\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}\right)$	BLANK1 No
$\mathcal{P}_3(\mathbb{R})$	$\mathcal{P}_3(\mathbb{R})$	$D(f) = f'$	$\mathcal{P}_0(\mathbb{R})$	BLANK2 No

injective iff $\text{null}(T) = \{0\}$

domain	codomain	T	$\text{null}(T)$	$\text{range}(T)$
$\mathbb{R}^3$ <i>dim = 3</i>	$\mathbb{R}$ <i>dim = 1</i>	$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$	<i>dim = 2</i> $\text{span}\left(\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}\right)$	$\mathbb{R}$ <i>dim = 1</i>
$\mathcal{P}_3(\mathbb{R})$ <i>dim = 4</i>	$\mathcal{P}_3(\mathbb{R})$ <i>dim = 4</i>	$T(f) = f'$	$\mathcal{P}_0(\mathbb{R})$ <i>dim = 1</i>	$\mathcal{P}_2(\mathbb{R})$ <i>dim = 3</i>

wooclap

What is $\dim(\text{null}(T)) + \dim(\text{range}(T))$? *dim domain*

11

Rank-Nullity Theorem. Let $\mathcal{V}$ be a finite-dimensional space.

Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a linear map. Then

$$\dim \text{null}(T) + \dim \text{range}(T) = \dim(\mathcal{V})$$

$\nearrow$ domain

$\downarrow$
aka nullity(T)

$\downarrow$
aka rank(T)

HBW week $\Rightarrow$ provide a proof

Linear Map Review

Q1. Let $T: \mathcal{V} \rightarrow \mathcal{W}$ be a function.

- a) Show that T is a linear map.
- b) Write down the matrix representation of T .
- c) Determine a basis for either $\text{null}(T)$ or $\text{range}(T)$.
- d) Determine $\dim \text{null}(T)$ and $\dim \text{range}(T)$.
- e) Determine whether
 - i.* T is injective;
 - ii.* T is surjective.